

SAMIA RIVERA

(929) 855-1439 • srive26@u.rochester.edu • [linkedin.com/in/samia-Rivera](https://www.linkedin.com/in/samia-Rivera) • srive26-design.github.io

EDUCATION:

University of Rochester — Rochester, NY

B.S. Electrical & Computer Engineering | Expected May 2028

GPA: 3.45

Relevant Coursework: Mechatronics and Embedded Systems, Digital Logic Design, Computer Organization, Signals, Circuits, Probability, Electronic Devices, Semiconductors

SKILLS:

Programming: C/C++, Verilog/VHDL, HTML, CSS | **Engineering Tools:** Onshape (3D CAD), Soldering, Microsoft Excel

Technical: Circuit Analysis, System-Level Debugging, Hardware-Software Integration

WORK EXPERIENCE:

Primergy — Interconnection & Transmission

Intern | Jun 2026-Present

- Rebuild and maintain interconnection process trackers in Excel across 3 different utilities, translating tariffs and business practices into standardized milestones, payment requirements, risks, and source-linked documentation
- Develop Salesforce workflow improvements with the Engineering and Development teams, translating team needs into business requirements that reduce duplicate data entry, link project records, and prioritize time-sensitive reviews

Department of Public Safety — University of Rochester

Public Safety Patrol Officer | Oct 2026-May 2026

- Patrolled residential areas during overnight shifts, monitored for safety concerns, and responded using established procedures.
- Communicated with supervisors and community members and documented incidents as needed.

Summer Youth Employment Program — CASW Inc.

Community Basketball Tournament Assistant | Jun 2025–Aug 2025

- Managed weekend setup and teardown for a summer-long community basketball tournament and reorganized the equipment shed, reducing setup time by 20%.
- Helped fund local basketball leagues through snack sales and handled tournament cash accurately.

Incoming Students Summer Program — Saint Catharine Academy

Summer Program Instructor | May 2023–Aug 2023

- Led hands-on STEM workshops for 20+ students, translating engineering concepts into interactive activities.
- Maintained a safe, structured classroom by enforcing procedures and addressing student needs.

Summer Youth Employment Program — Bronx Works

Applied Research Assistant | Jun 2022–Aug 2022

- Communicated technical findings and design constraints through presentations to program stakeholders.
- Researched and prototyped energy optimization and water purification concepts, evaluating efficiency, feasibility, and design tradeoffs

LEADERSHIP & COMMUNITY:

ECE 101 — University of Rochester

Workshop Leader | Sep 2026–Present

- Facilitate weekly workshops for six first-year ECE students, helping them work through course material and prepare for exams.
- Lead group activities and discussions that give students room to ask questions and learn from one another.

VEX Robotics Competition — Team 9717A

Lead Mechanic & Lead CAD Designer | Sep 2021–May 2024

- Led mechanical design and CAD development in Onshape, integrating drivetrain and manipulator systems while balancing performance and reliability.
- Built and tested mechanism prototypes and mentored 8+ students in CAD, mechanical design, and shop safety.

TECHNICAL PROJECTS:

Lunarbotics STEM Outreach Robotic Arm

Team Project | Sep 2026–Present

- Collaborate on a battery-powered Arduino Mega robotic arm planned for school STEM outreach demonstrations.
- Work with the team to specify candy-handling reach, payload, torque, power, and joint constraints before CAD and assembly; plan Xbox controller communication with the microcontroller.

Arduino Mega 2560 Microcontroller Labs

Course Projects | Sep 2026–Present

- Programmed a quadrature encoder using pin-change interrupts and Timer1 to report position and angle through UART; tested output-shaft rotations.
- Built Timer1 LED waveforms, a serial-controlled Morse code program, and an assembly timing loop on the ATmega2560.

C++ Console Game

Personal Project | Sep 2025–Present

- Develop a modular C++ console game with procedural maze generation, structured game loops, player state management, and input handling
- Build gameplay systems including minigames, HUD displays, and text-based animation while documenting development through a public YouTube showcase (750+ views, 26 comments)

Nintendo 3DS System Configuration & Firmware Exploration

Independent Technical Project | Apr 2025–Sep 2025

- Configured custom firmware and homebrew on Nintendo 3DS hardware, investigating boot processes and system behavior.
- Used low-level troubleshooting and documentation to debug system issues and examine hardware-software interaction.

Honors & Awards: VEX Robotics Regional Design Award, VEX Robotics State Judges Award x2, Dean's List, BridgeUP Scholar

Clubs: Lunar Botics, Society of Hispanic Professional Engineers, Women's Ultimate Frisbee, SWE

Interests: Robotics, Renewable Energy Systems, FPGA Design, Power Infrastructure, Baseball

Languages: English (Fluent), Spanish (Fluent), ASL (Intermediate)
